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Roll No

CM-7005(2)-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Polymer Technology

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.
iii) Assume suitable data if required and mention it clearly.
iv) Draw neat and clean diagrams as and when required to support your answer.

1. a) Give the differences between Cationic Polymerization and anionic Polymerization. 2
b) Explain the Cationic Polymerization in detail. 6
c) Explain the group transfer polymerization in detail. 6
2. a) Write short note on following polymerization techniques: 8
i) Bulk Polymerization Technique
ii) Solution Polymerization Technique
b) Write short note on thermoplastic composites. 6
3. a) Explain acidolysis, aminolysis and substitution reactions for polymers. 6
b) Write down various reactions for hydroxyl groups and aldehyde groups. 8
4. a) Write down manufacturing process and applications of Polyethylene. 7

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- b) Write short note on Cellulosic (Rayon) and Polyesters (Dacron) in detail. 7
- 5. a) Explain the ecology and environmental aspects of polymer industries. 7
- b) List out the polymer degradation types. Explain any two in detail. 7
- 6. a) Explain the polymerization kinetics for anionic polymerization. 7
- b) Write short note on fillers and additives for reinforced treatment. 7
- 7. a) Write down various reactions which lead to block copolymer. 7
- b) Write down manufacturing process and applications of epoxides. 7
- 8. a) Write short note on Acrylic-Olefin. 7
- b) Write short note on Uni-coelastic models- Maxwell. 7
